

SECTION 3.7: IMPROPER INTEGRALS (DAY 2)

Compute these integrals with friends! Please carefully write the limit, for example

$$\int_1^\infty \frac{dx}{x^2} = \lim_{t \rightarrow \infty} \int_1^t \frac{dx}{x^2} = \lim_{t \rightarrow \infty} \left[-\frac{1}{x} \right]_1^t = \lim_{t \rightarrow \infty} 1 - \frac{1}{t} = 1$$

$$1. \int_2^\infty \frac{1}{9+x^2} dx =$$

$$2. \int_{-\infty}^0 e^x dx =$$

$$3. \int_0^1 \frac{1}{\sqrt[4]{x}} dx =$$

$$4. \int_0^1 \ln t \, dt =$$

$$5. \int_1^2 \frac{dx}{1-x} =$$

$$6. \int_0^\infty e^x e^{-sx} dx =$$

$$7. \int_0^\pi \tan x dx =$$

$$8. \int_2^\infty \frac{dx}{x \ln^3 x} =$$